

LIGHTER

Institutional Research Memo

Perpetual Futures Exchange | Ethereum-Settled zk-Rollup

Date: 2025-12-24

Classification: Research-Only

Memo structure guided by the Underwriting Framework & Deep Research Stress-Test Guide and the Institutional-Grade Crypto Research Template v3.

HEADER

- **Stance:** Research-Only
- **Final Score:** 67 / 100
- **Sizing:** 0% today; would only size small/satellite post-tokenomics + governance clarity (see "Evidence-linked sizing logic").
- **Asset type:** Perps (exchange)
- **Chain / Stack:** Ethereum-settled custom zk-rollup (L2) with a centralized sequencer + ZK-enforced rules.
- **Liquidity regime (today):** Reflexive / Mercenary (points program + fee model suggest incentive-driven flow) [Inference].

EVIDENCE STRENGTH MAP

- **T1 sources used:** Lighter Docs (architecture; orders/matching), Blockworks (mainnet + architecture), L2BEAT (upgrade keys + sequencer + escape hatch), Fortune interview via Blockhead (funding/traction + founder bio). [Fact – T1]
- **T2 sources used:** Bitget News, NFTGators, Coinfomania (token notes / competitive volume claims / profitability claim). [Fact – T2]
- **Direct checks:** None performed in this memo (no independent on-chain queries); relying on the provided Evidence Log + red flags/open questions. [Unknown]

Evidence tag legend:

- [Fact – T1] = verified from Tier 1 / on-chain / official
- [Inference] = logical conclusion from facts
- [Estimate] = partial data / comps / back-of-envelope
- [Unknown] = explicitly not known or undisclosed

Key unverified but important items:

- LIT launch plan: allocations, vesting, utility, distribution mechanics. [Unknown]
- Decentralization timeline: sequencer + upgrade authority handoff. [Unknown]
- Insurance fund / "LLP" mechanics: deployed? coverage? insolvency handling. [Unknown]
- Audit scope / security attestations for custom circuits/prover. [Unknown]

TLDR

- **What the project is:** Lighter is a perpetual futures exchange implemented as an Ethereum-settled zk-rollup, with ZK proofs enforcing matching + liquidation rules. [Fact – T1]
- **Protocol mechanics snapshot (1 line, concrete):** Users deposit collateral → place perp orders → centralized sequencer batches → ZK proof enforces matching/liquidations on Ethereum → escape hatch allows withdrawals if sequencer stalls. [Fact – T1]
- **Why it matters now:** Public mainnet opened (Oct 2025) and adoption scaled quickly (users/TVL/OI + large daily volume reported) alongside a \$68M raise at \$1.5B valuation. [Fact – T1]
- **Why this research exists now:** Token (LIT) is minted but not launched and a points program is running; governance remains high-trust / centralized (Stage 0-style assumptions). [Fact – T1/T2 + Inference]

• **My current stance + size:** Research-Only; 0% until tokenomics + governance + insurance backstop disclosures improve. [Inference]

SCOREBOARD – 5 PILLARS (0 to 20 each)

- **Team and Execution:** 17 / 20
- **Technology, Security Model & Auditability:** 16 / 20
- **Tokenomics, Incentives & Liquidity Structure:** 8 / 20
- **Market Positioning, Narrative & Reflexivity:** 17 / 20
- **Risk Management, Governance & Attack Surface:** 9 / 20
- **Composite Score:** 67 / 100

Score Justification Matrix

Pillar	Key Drivers (why score is high)	Key Penalties (why not higher)	Net Outcome (1 line)
Team and Execution	Founder pedigree; shipped public mainnet; rapid traction metrics. [Fact – T1]	Team retention claim is T2; profitability claim is T2; limited disclosure on org + ops. [Fact – T2/Unknown]	Strong builder/story + traction, but incomplete institutional transparency.
Technology, Security Model & Auditability	ZK-enforced matching/ liquidations; custom proving engine; stated price-time priority enforcement. [Fact – T1]	Centralized sequencer; early closed-source circuits; audit scope not described. [Fact – T1 + Unknown]	Technically compelling, but trust assumptions + verification surface remain.
Tokenomics, Incentives & Liquidity	Fixed supply token minted; active points program to drive activity. [Fact – T1/T2]	No TGE/distribution; allocations/vesting/utility unclear; potential insider concentration risk. [Unknown]	Token remains the largest diligence gap; caps conviction.
Market Positioning, Narrative & Reflexivity	\$68M raise at \$1.5B; fee-free retail + paid pro access; large reported volume/TVL/OI/user growth. [Fact – T1]	"Beats Hyperliquid 30-day volume" is T2; revenue quality not disclosed. [Fact – T2/Unknown]	Clear momentum + narrative flywheel; some key comps rely on T2.
Risk, Governance & Attack Surface	Escape hatch / "Desert Mode" mitigates liveness risk. [Fact – T1]	Upgrades controlled by small multisigs (timelock bypass); single sequencer MEV/ censorship risk; insurance fund unclear. [Fact – T1/Unknown]	Governance + centralization are the primary institutional blockers today.

COMPETITIVE CONTEXT

Direct competitors:

- Hyperliquid – benchmark perp DEX competitor; volume comparison cited as near/above. [Fact – T2]
- Aster – cited as overtaken on 24h volume (Nov 2025). [Fact – T1]

Adjacent competitors:

- Other decentralized perp exchanges (not enumerated in Evidence Log). [Unknown]

Why this market matters now:

- Lighter is reporting very high perp volumes (e.g., ~\$10.7B/24h on a cited day) plus rapid TVL/OI/user growth, implying real demand for high-performance on-chain perps. [Fact – T1]
- The differentiation attempt is "exchange performance + verifiability" (ZK-enforced rules) rather than purely incentive spend. [Fact – T1 + Inference]

Liquidity structure & depth regime:

- Market type: Orderbook / matching engine with price-time priority claims. [Fact – T1]
- Depth snapshot: Not disclosed (no slippage / depth metrics provided). [Unknown]
- LP concentration / top holders: N/A / not provided (orderbook venue; token not circulating). [Unknown]

Where this project sits vs peers:

- Positioned as a top challenger in decentralized perps; one source claims 30-day volume slightly above Hyperliquid late-2025. [Fact – T2]
- Differentiators: fee-free retail (UI), paid low-latency for pros, and ZK-enforced exchange rules. [Fact – T1/T2]

Dominant competitive drivers:

- Liquidity depth and velocity [Unknown]
- Execution quality + latency segmentation (retail vs pro) [Fact – T2]
- Governance clarity relative to peers (currently weak) [Fact – T1]
- Audit coverage + open-source verification surface [Unknown]
- Narrative reflexivity: founder story + high-status investors [Fact – T1]

Competitive reflexivity hooks:

- Points program + fee-free retail can amplify volume and social proof. [Inference]
- "High-status backers" narrative can accelerate onboarding and liquidity migration. [Inference]

TEAM AND EXECUTION – 17 / 20

- **Team visibility and credibility:** Founder is publicly profiled with exceptional background (Harvard early; Citadel). [Fact – T1]
- **Execution proof:** Public mainnet launch + rapid scaling in users/TVL/OI/volume; major Series A financing. [Fact – T1]
- **Accountability and communication:** Not enough incident history / postmortems / operational transparency in Evidence Log. [Unknown]

Key strengths:

- Founder pedigree and credible "builder" narrative. [Fact – T1]
- Shipped a differentiated stack (zk-rollup perps) + got users. [Fact – T1]
- Strong institutional investor syndicate signals diligence. [Fact – T1]

Key weaknesses or unknowns:

- Team continuity claim (~80% retained) is Tier 2. [Fact – T2]
- Profitability claim exists but is Tier 2 and not quantified. [Fact – T2/Unknown]
- Limited disclosures on operations, security process, and accountability. [Unknown]

TECHNOLOGY, SECURITY MODEL & AUDITABILITY – 16 / 20

- **Architecture in simple language:** Ethereum-anchored zk-rollup where a centralized sequencer batches trades, but ZK proofs validate matching/liquidations on Ethereum. [Fact – T1]
- **Inherited modules and dependencies:** Ethereum settlement + custom ZK circuits; fast USDC bridging mentioned (CCTP integration referenced in appendix). [Fact – T1 + Unknown (independent verification)]
- **Audit status and coverage:** Not detailed (named auditors / scope not provided). [Unknown]

Key strengths:

- ZK-enforced rule compliance for exchange state transitions. [Fact – T1]
- Proprietary proving engine claimed to be built in-house for throughput. [Fact – T1]
- Matching rules claim "price-time priority enforced in circuit." [Fact – T1]

Key weaknesses or unknowns:

- Centralized sequencer remains a MEV/censorship vector. [Fact – T1]
- Early custom circuits/matching logic were initially closed-source. [Fact – T1 (from red flag section)]
- Proof engine/circuit audits, formal verification, and bug bounty details are missing. [Unknown]

TOKENOMICS, INCENTIVES & LIQUIDITY STRUCTURE – 8 / 20

- **Token roles:** LIT is described as intended for governance/incentives; exact role set is not specified. [Fact – T2 + Unknown]
- **Allocation and emissions:** Not disclosed (no allocation / vesting schedules provided). [Unknown]
- **Incentive structure and sustainability:** Active points program (Season 2) through end-2025, widely interpreted as a precursor to token distribution (not confirmed). [Fact – T1 (program) + Inference (airdrop expectation)]
- **Reflexivity and treasury dependencies:** Not disclosed. [Unknown]
- **Liquidity structure:** Token not live/circulating per Evidence Log; no CEX/DEX liquidity disclosed. [Unknown]
- **Distribution and fairness notes:** Evidence Log states fixed supply minted; one source claims deployer initially held full supply. [Fact – T2]

Key strengths:

- Fixed supply is defined (1B minted) even if launch is pending. [Fact – T2]
- Points program provides a potential distribution funnel. [Fact – T1 + Inference]

Key weaknesses or unknowns:

- Token launch details are the core gap: allocations, vesting, and utility not disclosed. [Unknown]
- Insider concentration and sell-pressure risk cannot be assessed today. [Unknown]

MARKET POSITIONING, NARRATIVE & REFLEXIVITY – 17 / 20

- **Category and role:** High-performance decentralized perps exchange implemented as an Ethereum-settled zk-rollup. [Fact – T1]
- **Narrative hooks:** "Verifiable fairness" (ZK enforced matching/liquidations), fee-free retail, high-status investors, founder-genius storyline. [Fact – T1/T2 + Inference]
- **Competitor advantage or weakness:** One source claims 30-day volume slightly surpassing Hyperliquid late-2025 (T2). [Fact – T2]
- **Liquidity flows and timing:** Fee segmentation (0 fees retail; small fees for low-latency pros) + points program can drive activity. [Fact – T2/T1 + Inference]
- **Sentiment and narrative reflexivity:** Major financing round and investor roster likely reinforce adoption flywheel. [Inference]

Key strengths:

- Strong "institutional signal" from a \$68M round at \$1.5B valuation. [Fact – T1]
- Rapid adoption metrics suggest meaningful early product-market pull. [Fact – T1]

Key weaknesses or unknowns:

- Some "#1 vs peers" metrics rely on Tier 2 reporting. [Fact – T2]
- Sustainability of volumes vs incentive/reflexive dynamics is not established. [Unknown]

RISK MANAGEMENT, GOVERNANCE & ATTACK SURFACE – 9 / 20

- **Governance model:** Upgradable contracts controlled by small multisigs; "network governor" multisig with timelock, and "security council" multisig can bypass delay. [Fact – T1]
- **Key control and multisig:** 2-of-3 (network governor) and 3-of-5 (security council). [Fact – T1]
- **Upgrade paths and emergency powers:** Nominal 21-day delay, but bypass exists for emergencies via council. [Fact – T1]
- **Treasury governance:** Not described. [Unknown]
- **Audit coverage and security track record:** Not detailed; early reliance on audits noted but specifics missing. [Unknown]
- **Early stage fragility:** Single sequencer and centralized upgrades are primary trust assumptions. [Fact – T1]

Oracle & dependency map:

- Oracles: Not specified. [Unknown]
- Bridges: Fast USDC bridge / CCTP integration mentioned in appendix; not independently verified here. [Unknown]
- L2 / sequencer dependence: Centralized sequencer; operator-level MEV/censorship risk. [Fact – T1]
- External contracts: Not specified beyond listed components. [Unknown]

Security model reality check:

- Upgradeability: Contracts are upgradable; small multisigs can change code. [Fact – T1]
- Admin keys / multisigs: Identified (2-of-3 and 3-of-5). [Fact – T1]
- Backdoors / god modes: Emergency upgrade bypass exists; broader "pause/seize" powers not specified in Evidence Log. [Fact – T1 + Unknown]

Attack surface map (technical/economic):

- Technical: Custom ZK circuits/prover complexity risk; upgrade risk; bridge complexity (if used). [Inference]
- Economic: Sequencer-level MEV/censorship; insolvency/large liquidation loss handling depends on insurance design (unclear). [Fact – T1 + Unknown]
- Stress points and complexity risks: Exchange-grade throughput + proving system + rollup upgrades create clustered tail risk. [Inference]

Thesis-breaking risks (kill-shots):

- Multisig compromise or abuse → hostile upgrade / loss of trust. [Inference based on governance facts]
- Sequencer misbehavior/outage beyond tolerance → forced halts/withdrawals and liquidity flight. [Fact – T1 + Inference]
- Token launch reveals misaligned allocations / heavy insider overhang. [Unknown]
- Insurance/LLP proves insufficient in extreme liquidation event. [Unknown]

Attack surface map (governance / social):

- Governance: Centralization and timelock bypass weaken credible neutrality. [Fact – T1]
- Social: Narrative-driven flows could reverse if tokenomics or governance disappoint. [Inference]

Governance failure paths (explicit):

- If 2-of-3 or 3-of-5 signers go rogue, plausible outcome is immediate hostile upgrade or rule changes. [Inference]

- If the timelock bypass is used non-transparently, plausible outcome is loss of user trust → liquidity exit. [Inference]

CONTRADICTION LOG

- "Price-time-priority is mathematically enforced; no malicious reordering" (Docs, T1) vs "Sequencer can insert own transactions / MEV without user protection" (L2BEAT, T1) – stronger source: Both T1; unresolved scope conflict.
- "Already profitable" (NFTGators, T2) vs lack of disclosed revenue/financials – stronger source: Unknown (needs T1 financial disclosure).

VERDICT

- Lighter looks like a credible, fast-moving on-chain perps challenger with a differentiated technical angle (zk-enforced exchange rules) and strong early traction. [Fact – T1]
- However, the project remains high-trust: centralized sequencer and concentrated upgrade authority materially increase governance and tail-risk. [Fact – T1]
- Token readiness is the biggest unknown: LIT is minted but distribution/utility/vesting are unclear, limiting institutional position sizing today. [Fact – T2 + Unknown]
- Stance consistent with this: Research-Only, pending disclosures and decentralization milestones. [Inference]

Evidence-linked sizing logic

What pushes sizing up:

- Published, credible LIT tokenomics (allocations, vesting, utility) with alignment. [Unknown → would become Fact]
- Clear decentralization roadmap executed (sequencer + upgrades). [Unknown → would become Fact]
- Independent security assurance on custom circuits/prover (scope + findings). [Unknown → would become Fact]
- Evidence of sustained activity beyond points-driven flow (retention, stable OI/TVL). [Unknown]

What keeps sizing capped or pushes sizing down:

- Continued centralized control with timelock bypass as default posture. [Fact – T1]
- Token launch with opaque or insider-heavy distribution. [Unknown]
- Any security/upgrade incident or credibility shock around "verifiable fairness." [Unknown]

UPSIDE CASE

- ZK-enforced matching becomes a trusted "fair execution" standard. [Inference]
- Volume/TVL/OI remain strong post-incentives; retention improves. [Unknown]
- Governance decentralizes; trust assumptions compress materially. [Unknown]
- Token launch aligns users/operators without heavy sell overhang. [Unknown]

RISK CASE

- Multisig or upgrade pathway risk triggers loss of trust and exit. [Inference]
- Sequencer-level MEV/censorship narrative becomes credible → liquidity migration. [Fact – T1 + Inference]
- Tokenomics disappoint (distribution/utility) and reflexivity reverses. [Unknown]
- Insurance/LLP is inadequate in extreme liquidation events. [Unknown]

SCENARIO MAP – BULL, BASE, BEAR

Bull case

Lighter sustains leadership-level perps activity, proves reliable execution and economics, and meaningfully decentralizes sequencer + upgrades, reducing "Stage 0" trust assumptions. [Inference]

Bull case KPIs (on-chain where possible):

- Sustained high volume vs 2025 peak days (not just one-off spikes). [Unknown]
- Governance milestones: reduced multisig power / timelock bypass constraints. [Unknown]
- Public security artifacts: audit scope, remediation evidence. [Unknown]

Base case

Lighter continues operating with strong product momentum, but remains centralized for longer; token launch remains delayed or launches with moderate alignment; growth continues but with higher risk discount. [Inference]

Base case KPIs:

- TVL/OI stabilize near current reported ranges rather than compounding. [Unknown]
- Partial governance upgrades (better transparency but still concentrated keys). [Unknown]

Bear case

A governance/security shock or tokenomics disappointment triggers a confidence break: volumes/TVL fall, and the system's centralized trust assumptions become the focal point of a liquidity exit. [Inference]

Bear case KPIs:

- Sharp declines in reported TVL/OI and daily volume from late-2025 levels. [Unknown]
- Emergency upgrades / frequent "hotfix" patterns that erode trust. [Unknown]
- Evidence of escape hatch usage ("Desert Mode") rising during stress. [Unknown]

WHAT I AM WATCHING NEXT

- LIT token launch documentation (allocations/vesting/utility). [Unknown]
- Sequencer decentralization plan and upgrade authority reductions. [Unknown]
- Security posture: audits, public repos/circuit transparency, proofs verification surface. [Unknown]
- Sustainability of traction metrics after incentive seasons change. [Unknown]
- Insurance fund / LLP design disclosure and stress-case handling. [Unknown]

UNKNOWN VARIABLES THAT MOVE CONVICTION (CRYPTO-SPECIFIC)

- Real unit economics: fees from pro traders, margins, profitability evidence. [Unknown]
- Token distribution fairness and sell-pressure risk at launch. [Unknown]
- Sequencer MEV mitigation beyond "matching fairness" claims. [Unknown]
- Insurance/LLP existence, rules, and sufficiency under extreme liquidations. [Unknown]

BIAS, POV & TIME SENSITIVITY

- **My bias / POV:** I may overweight "technically elegant" designs (zk-fairness) relative to governance centralization risks. [Inference]
- **What new info would most change my stance:** tokenomics docs, decentralization execution, and credible security attestations. [Unknown]

Time sensitivity notes:

- Fast decaying findings: daily volume spikes, incentives/points dynamics, short-term narrative dominance. [Inference]
- Slow changing structural points: zk-rollup architecture, upgrade model (unless changed), trust assumptions. [Fact – T1 + Inference]
- Items that require periodic review: governance key changes, token distribution readiness, TVL/OI stability, incident history. [Unknown]

Not financial advice. Research only; execution depends on liquidity, flow and market structure.